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WATER PUMP DIAGNOSIS DEVICE

Abstract

A water pump diagnosis device diagnoses a water pump that circulates a coolant that cools a cooling target. The water pump diagnosis device includes a temperature sensor and a diagnosis unit. The temperature sensor measures a temperature of a power semiconductor, which is a part of the cooling target. The diagnosis unit determines whether the water pump has failed based on a comparison between a temperature rise rate of the power semiconductor based on a temperature measured by the temperature sensor and a preset reference temperature rise rate.

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Background/Summary

[0001] This application is based on and claims the benefit of priority from Japanese Patent Application No. 2024-034273, filed on 6 Mar. 2024, the content of which is incorporated herein by reference.

BACKGROUND OF THE INVENTION

Field of the Invention

[0002] The present invention relates to a water pump diagnosis device for detecting a failure of a water pump.

Related Art

[0003] Some vehicles include a water pump that circulates a coolant for cooling a cooling target such as a vehicle control unit.

[0004] Patent Document 1: Japanese Unexamined Patent Application, Publication No. 2017-115680

SUMMARY OF THE INVENTION

[0005] When the water pump fails and the flow rate of the coolant decreases, various problems may occur due to a rise in the temperature of the cooling target. Therefore, it is preferable for it to be possible to quickly diagnose whether or not the water pump has failed.

[0006] As a technique of detecting a failure of a water pump, for example, there is a technique of diagnosing whether or not the water pump has a failure based on the rotation speed of the water pump. However, with such a method, it is not possible to diagnose whether or not a water pump to which the rotation speed sensor is not attached has failed.

[0007] The present invention has been made in view of the above circumstances, and it is an object of the present invention to diagnose whether or not a water pump has failed without being based on the rotation speed of the water pump.

[0008] The inventors of the present invention have found that it is possible to detect the failure of a water pump without being based on the rotation speed of the water pump, when focusing on a temperature rise of a power semiconductor as a part of a cooling target, thereby arriving at the present invention. The present invention provides the following water pump diagnostic devices (1) to (5). [0009] (1) A water pump diagnosis device for diagnosing a water pump that circulates a coolant for cooling a cooling target, the water pump diagnosis device including:

a temperature sensor that measures a temperature of a power semiconductor, which is a part of the cooling target; and a diagnoser that determines whether the water pump has failed based on a comparison between a temperature rise rate of the power semiconductor based on a temperature measured by the temperature sensor and a preset reference temperature rise rate.

[0010] According to the present configuration, it is possible to estimate the flow rate of the coolant from the comparison between the temperature rise rate of the power semiconductor and the reference temperature rise rate. That is, it is possible to estimate that the flow rate of the coolant is smaller as the temperature rise rate of the power semiconductor is higher than the reference temperature rise rate. Therefore, it is possible to diagnose whether or not the water pump has failed without being based on the rotation speed of the water pump. [0011] (2) In the water pump diagnosis device according to (1), the cooling target, the water pump, and the water pump diagnosis device are installed in a vehicle,

the cooling target includes a power module for driving the vehicle, and

the power semiconductor is a part of the power module.

[0012] According to the present configuration, in the vehicle, it is possible to obtain an advantageous effect of diagnosing whether or not the water pump has failed without being based on the rotation speed of the water pump. [0013] (3) The water pump diagnosis device as described in (1) or (2) further includes an alarm that warns that the water pump has failed on a condition that the diagnoser determines that the water pump has failed.

[0014] According to the present configuration, it is possible for the driver to quickly recognize the

failure when the water pump fails. [0015] (4) In the water pump diagnosis device as described in any one of (1) to (3), the diagnoser
determines that the water pump has failed on a condition that a rise rate difference defined as a value obtained by subtracting the reference temperature rise rate from the temperature rise rate of the power semiconductor based on a temperature measured by the temperature sensor is larger than a predetermined failure determination value, and
determines that the water pump has stopped on a condition that the rise rate difference is larger than a stop determination value which is larger than the failure determination value.
[0016] According to the present configuration, it is possible to diagnose in detail the state of the water pump. [0017] (5) The water pump diagnosis device as described in (4), further includes an alarm that warns that the water pump has failed on a condition that the diagnoser determines that the water pump has failed, and warns that the water pump has stopped on a condition that the diagnoser determines that the water pump has stopped.
[0018] According to the present configuration, it is possible for the driver to recognize in detail the state of the water pump.
[0019] As described above, according to the configuration (1), it is possible to diagnose whether or not the water pump has failed without being based on the rotation speed of the water pump. Further, according to the configurations (2) to (4) citing (1), it is possible to obtain the respective additional advantageous effects.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0020] FIG. **1** is a configuration diagram showing a water pump diagnostic device and its periphery according to a first embodiment;
[0021] FIG. **2** is a plan view showing an example of a power semiconductor and its periphery;
[0022] FIG. **3** is a front view showing an example of a power semiconductor and its periphery;
[0023] FIG. **4** is a graph showing a relationship between a working time and a temperature of a power semiconductor for every flow rate of a coolant; and
[0024] FIG. **5** is a flowchart showing a flow of diagnosis by the water pump diagnosis device.
DETAILED DESCRIPTION OF THE INVENTION
[0025] Hereinafter, embodiments of the present invention will be described with reference to the drawings. However, the present invention is not limited to the following embodiments, and can be appropriately modified and implemented within a range not departing from the gist of the present invention.
First Embodiment
[0026] As shown in FIG. **1**, a water pump diagnosis device **60** of the present embodiment is installed in a vehicle **100**. The vehicle **100** includes a VCU **70**, a water pump **80**, and a radiator **90**. The VCU is an abbreviation for “Vehicle Control Unit”.
[0027] The VCU **70** includes an ECU **20**, a capacitor **30**, an IPM **40**, and a reactor **50**. Note that “ECU” is an abbreviation for “Electronic Control Unit”. Further, “IPM” is an abbreviation for “Intelligent Power Module”. That is, the “IPM” is a power module for driving the vehicle.
[0028] The IPM **40** includes an input-side circuit for inputting power to the reactor **50** and an output-side circuit for outputting power from the reactor **50**. Therefore, as shown in FIG. **2**, the IPM **40** includes a plurality of power semiconductors such as a semiconductor switch **44** and a diode **45**. The capacitor **30** shown in FIG. **1** is, for example, a smoothing capacitor in the input-side circuit or a smoothing capacitor in the output-side circuit.
[0029] The ECU **20** shown in FIG. **1** controls the IPM **40** by controlling the semiconductor switches **44** shown in FIG. **2**.

[0030] The water pump **80** shown in FIG. **1** circulates a coolant CL for cooling the condenser **30**, the IPM **40**, and the reactor **50** in the vehicle **100**. Hereinafter, the capacitor **30**, the IPM **40**, and the reactor **50** will be referred to as “cooling targets **30**, **40**, **50**”. The water pump **80** cools the cooling targets **30**, **40**, and **50** by circulating the coolant CL between the cooling targets **30**, **40**, and **50** and the radiator **90**. Specifically, for example, as illustrated in FIG. **3**, the coolant CL passes between fins **48** for promoting cooling below the cooling targets **30**, **40**, and **50**.
[0031] As shown in FIG. **1**, the water pump diagnosis device **60** includes a temperature sensor **61**, a temperature rise rate identification unit **62**, a diagnosis unit **63**, and a warning unit **65**. The temperature sensor **61** is installed in the IPM **40**. Each of the temperature rise rate identification unit **62** and the diagnosis unit **63** is configured as a part of the ECU **20**. The warning unit **65** is provided in front of or in the vicinity of the driver's seat in the vehicle **100**.
[0032] The temperature sensor **61** measures the temperature Ta of the semiconductor switch **44**. The temperature rise rate identification unit **62** identifies the temperature rise rate Va of the semiconductor switch **44** from the rise in the temperature Ta of the semiconductor switch **44** relative to the operation time t of the semiconductor switch **44**.
[0033] The diagnosis unit **63** stores a preset reference temperature rise rate Vb. Specifically, the diagnosis unit **63** maps and stores the reference temperature rise rate Vb for every situation. As shown in FIG. **4**, the reference temperature rise rate Vb is a rise rate of the temperature Ta=Tb of the semiconductor switch **44** when the water pump **80** is normal, that is, when the flow rate of the coolant CL is high. The reference temperature rise rate Vb may be, for example, a value obtained by experiment in advance or a value obtained by simulation in advance.
[0034] The diagnosis unit **63** shown in FIG. **1** diagnoses the water pump **80** based on a comparison between the temperature rise rate Va of the semiconductor switch **44** identified by the temperature rise rate identification unit **62** and the reference temperature rise rate Vb. Hereinafter, a value (Va-Vb) obtained by subtracting the reference temperature rise rate Vb from the temperature rise rate Va of the semiconductor switch **44** is referred to as a “rise rate difference ΔV”. The two predetermined threshold values are referred to as a “failure determination value th**1**” and a “stop determination value th**2**” in order from smaller.
[0035] When the rise rate difference ΔV is less than the failure determination value th**1**, the diagnosis unit **63** determines that the water pump **80** is normal. On the other hand, when the rise rate difference ΔV exceeds the failure determination value th**1**, the diagnosis unit **63** determines that the water pump **80** has failed. Further, when the rise rate difference ΔV exceeds the stop determination value th**2**, the diagnosis unit **63** determines that the water pump **80** has stopped.
[0036] The warning unit **65** warns the driver of the vehicle **100** that the water pump **80** has failed on condition that the diagnosis unit **63** has determined that the water pump **80** has failed. Specifically, for example, a warning that the flow rate of the coolant CL is decreasing is issued. The warning unit **65** warns the driver of the vehicle **100** that the water pump **80** has stopped on the condition that the diagnosis unit **63** determines that the water pump **80** has stopped. These warnings may be, for example, visually appealing such as a warning screen display or a warning lamp, may be audibly appealing such as a warning sound or a warning announcement, or may be both.
[0037] Next, the flow of diagnosis by the water pump diagnosis device **60** described above will be described with reference to FIG. **5**. In the following description, the “S” added before a numeral is an abbreviation for “step”.
[0038] First, in S**1**, the temperature sensor **61** detects the temperature Ta of the semiconductor switch **44**. Next, in S**2**, the temperature rise rate identification unit **62** identifies the temperature rise rate Va of the semiconductor switch **44**.
[0039] Next, in S**3**, the diagnosis unit **63** determines whether or not the rise rate difference ΔV has exceeded the failure determination value th**1**. When the diagnosis unit **63** makes a negative determination, that is, when the rise rate difference ΔV is less than the failure determination value th**1**, it is determined that the water pump **80** is normal, and the processing flow ends. On the other

hand, when the diagnosis unit **63** makes an affirmative determination in S**3**, that is, when the rise rate difference ΔV has exceeded the failure determination value th**1**, it is determined that the water pump **80** has failed, and the processing advances to S**4**.

[0040] In S**4**, the diagnosis unit **63** determines whether or not the rise rate difference ΔV has exceeded the stop determination value th**2**. When the diagnosis unit **63** makes a negative determination, that is, when the rise rate difference ΔV is less than the stop determination value th**2**, it is determined that the water pump **80** is not stopped, and the processing advances to S**5**. In S**5**, the warning unit **65** warns that the flow rate of the coolant is decreasing.

[0041] On the other hand, when the diagnosis unit **63** makes an affirmative determination in S**5** described above, that is, when the rise rate difference ΔV has exceeded the stop determination value th**2**, it is determined that the water pump **80** has stopped, and the processing advances to S**6**. In S**6**, the warning unit **65** warns that the water pump **80** has stopped.

[0042] The configuration and advantageous effects of the present embodiment will be summarized below.

[0043] The temperature sensor **61** shown in FIG. **1** measures the temperature Ta of the semiconductor switch **44**, which is a part of the cooling targets **30**, **40**, and **50**. The diagnosis unit **63** determines whether or not the water pump **80** has failed based on a comparison between the temperature rise rate Va of the semiconductor switch **44** based on the temperature Ta measured by the temperature sensor **61** and a preset reference temperature rise rate Vb. That is, it can be estimated that the flow rate of the coolant CL is smaller as the temperature rise rate Va of the semiconductor switch **44** is higher than the reference temperature rise rate Vb. Therefore, it is possible to diagnose whether or not the water pump **80** has failed without being based on the rotation speed of the water pump **80**.

[0044] The cooling targets **30**, **40**, **50**, the water pump **80**, and the water pump diagnosis device **60** are installed in the vehicle **100**. The cooling targets **30**, **40**, and **50** include an IPM **40** for driving the vehicle **100**. The semiconductor switch **44** for measuring the temperature is a part of the IPM **40**. Therefore, in the vehicle **100**, it is possible to obtain the above advantageous effect, that is, an advantageous effect of possible to diagnose whether or not the water pump **80** has failed without being based on the rotation speed of the water pump **80**.

[0045] The warning unit **65** warns the driver of the vehicle **100** that the water pump **80** has failed on condition that the diagnosis unit **63** has determined that the water pump **80** has failed. Therefore, it is possible for the driver to quickly recognize that the water pump **80** has failed.

[0046] The diagnosis unit **63** determines that the water pump **80** has failed on the condition that the rise rate difference ΔV is larger than the failure determination value th**1**. In addition, the diagnosis unit **63** determines that the water pump **80** has stopped on the condition that the rise rate difference ΔV is larger than the stop determination value th**2**. Therefore, it is possible to diagnose in detail the failure state of the water pump.

[0047] The warning unit **65** warns the driver that the water pump **80** has failed on condition that the diagnosis unit **63** determines that the water pump **80** has failed. Further, the warning unit **65** warns the driver that the water pump **80** has stopped on the condition that the diagnosis unit **63** determines that the water pump **80** has stopped. Therefore, it is possible for the driver to recognize in detail the failure state of the water pump **80**.

Other Embodiments

[0048] The embodiment described above can be modified as follows, for example.

[0049] When the rise rate difference ΔV is larger than the failure determination value th**1**, the diagnosis unit **63** may only determine that the water pump **80** has failed, and may not determine whether the water pump **80** has stopped. The warning unit **65** may only warn the driver that the water pump **80** has failed on condition that the diagnosis unit **63** determines that the water pump **80** has failed.

[0050] In a case in which the diagnosis unit **63** determines that the water pump **80** has failed or has stopped, a predetermined fail-safe action may be performed instead of or in addition to the warning by the warning unit **65**.

[0051] Instead of measuring the temperature of the semiconductor switch **44** shown in FIG. **2**, the temperature of the diode **45** may be measured by the temperature sensor **61**.

EXPLANATION OF REFERENCE NUMERALS

[0052] **44** Semiconductor Switch (Power Semiconductor) [0053] **45** Diode (Power Semiconductor) [0054] **60** Water Pump Diagnosis Device [0055] **61** Temperature Sensor [0056] **63** Diagnosis Unit [0057] **65** Warning Unit [0058] **80** Water Pump [0059] **100** Vehicle [0060] Th**1** Failure Determination Value [0061] Th**2** Stop Determination Value [0062] Va Temperature Rise Rate of Semiconductor Switch (Temperature [0063] Rise Rate of Power Semiconductor) [0064] Vb Reference Temperature Rise Rate [0065] ΔV Rise Rate Difference

Claims

1. A water pump diagnosis device for diagnosing a water pump that circulates a coolant for cooling a cooling target, the water pump diagnosis device comprising: a temperature sensor that measures a temperature of a power semiconductor, which is a part of the cooling target; and a diagnoser that determines whether the water pump has failed based on a comparison between a temperature rise rate of the power semiconductor based on a temperature measured by the temperature sensor and a preset reference temperature rise rate.

2. The water pump diagnosis device according to claim 1, wherein the cooling target, the water pump, and the water pump diagnosis device are installed in a vehicle, the cooling target includes a power module for driving the vehicle, and the power semiconductor is a part of the power module.

3. The water pump diagnosis device according to claim 1, further comprising an alarm that warns that the water pump has failed on a condition that the diagnoser determines that the water pump has failed.

4. The water pump diagnosis device according to claim 1, wherein the diagnoser determines that the water pump has failed on a condition that a rise rate difference defined as a value obtained by subtracting the reference temperature rise rate from the temperature rise rate of the power semiconductor based on a temperature measured by the temperature sensor is larger than a predetermined failure determination value, and determines that the water pump has stopped on a condition that the rise rate difference is larger than a stop determination value which is larger than the failure determination value.

5. The water pump diagnosis device according to claim 4, further comprising an alarm that warns that the water pump has failed on a condition that the diagnoser determines that the water pump has failed, and warns that the water pump has stopped on a condition that the diagnoser determines that the water pump has stopped.